G7: The training loss and what the adapter returns

k4nd weakness 4 (two parts) and question 4 · status: answered

THE REVIEWER'S POINT

k4nd, weakness 4

4. Essential method details are poorly presented and sometimes inconsistent.

Section 4 describes target matching and provides the teacher prediction and sampling rule, but it does not explicitly formulate the training loss. The squared-error objective appears in the training diagram in Appendix A, Figure 8. Essential information about the optimization should be stated clearly in the main method section.

There are also substantive ambiguities. The main text describes individual adapted network evaluations as returning the correction, whereas Appendix B defines them as adapted denoising predictions whose composition induces the correction. Section 4 refers to "Figure 10," although the actual training diagram is Figure 8.

Reviews.pdf, page 5 (no text layer; passage located by OCR)

K-W4a

K-W4a

K-W4b

k4nd, question 4

- Can the authors state the optimization objective explicitly in the main text and resolve the inconsistencies concerning network outputs, figure references, and baseline starting noise?

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K-Q4

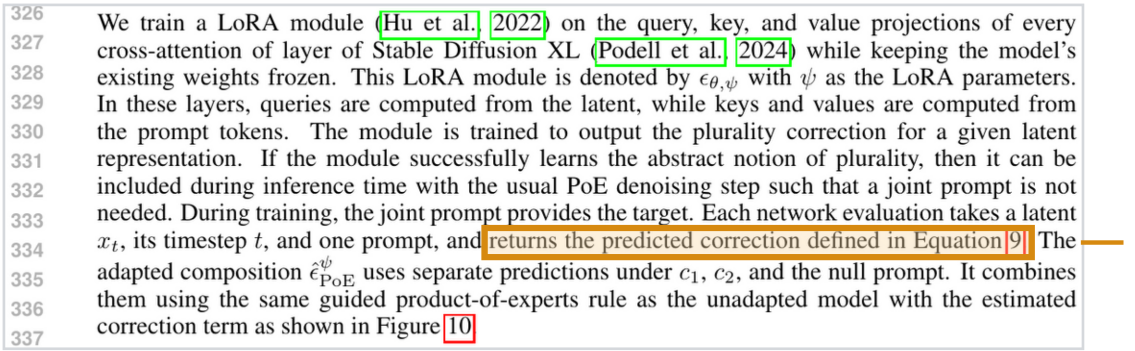
this sentence

no loss here

the diagram

THE SUBMITTED WEAKNESS

Section 4, opening paragraph



We train a LoRA module (Hu et al., 2022) on the query, key, and value projections of every cross-attention of layer of Stable Diffusion XL (Podell et al., 2024) while keeping the model's existing weights frozen. This LoRA module is denoted by $\epsilon_{\theta,\psi}$ with ψ as the LoRA parameters. In these layers, queries are computed from the latent, while keys and values are computed from the prompt tokens. The module is trained to output the plurality correction for a given latent representation. If the module successfully learns the abstract notion of plurality, then it can be included during inference time with the usual PoE denoising step such that a joint prompt is not needed. During training, the joint prompt provides the target. Each network evaluation takes a latent x_t, its timestep t, and one prompt, and returns the predicted correction defined in Equation 9. The adapted composition $\hat{\epsilon}^{\psi}_{\text{PoE}}$ uses separate predictions under c_1, c_2, and the null prompt. It combines them using the same guided product-of-experts rule as the unadapted model with the estimated correction term as shown in Figure 10

56173_Restoring_plurality_in_c.pdf (submitted), page 7

returns the correction

Section 4, how the adapter is trained

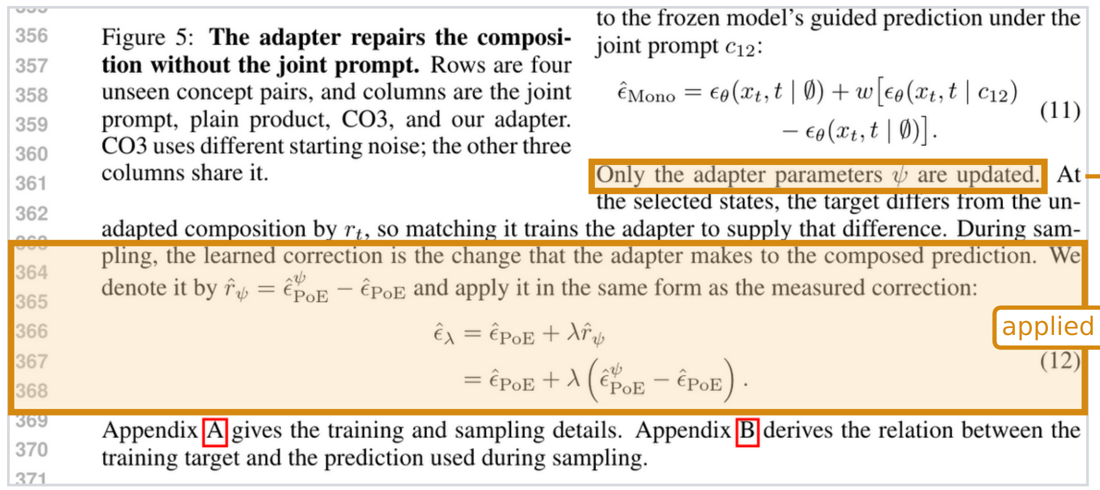


Figure 5: **The adapter repairs the composition without the joint prompt.** Rows are four unseen concept pairs, and columns are the joint prompt, plain product, CO3, and our adapter. CO3 uses different starting noise; the other three columns share it.

to the frozen model's guided prediction under the joint prompt c_{12}:

$$\hat{\epsilon}_{\text{Mono}} = \epsilon_\theta(x_t, t \mid \emptyset) + w\big[\epsilon_\theta(x_t, t \mid c_{12}) - \epsilon_\theta(x_t, t \mid \emptyset)\big]. \tag{11}$$

Only the adapter parameters ψ are updated. At the selected states, the target differs from the unadapted composition by r_t, so matching it trains the adapter to supply that difference. During sampling, the learned correction is the change that the adapter makes to the composed prediction. We denote it by $\hat{r}_\psi = \hat{\epsilon}^{\psi}_{\text{PoE}} - \hat{\epsilon}_{\text{PoE}}$ and apply it in the same form as the measured correction:

$$\begin{aligned}\hat{\epsilon}_\lambda &= \hat{\epsilon}_{\text{PoE}} + \lambda \hat{r}_\psi \\ &= \hat{\epsilon}_{\text{PoE}} + \lambda\left(\hat{\epsilon}^{\psi}_{\text{PoE}} - \hat{\epsilon}_{\text{PoE}}\right).\end{aligned} \tag{12}$$

Appendix A gives the training and sampling details. Appendix B derives the relation between the training target and the prediction used during sampling.

56173_Restoring_plurality_in_c.pdf (submitted), page 7

no loss written

applied as a correction at strength λ

Appendix A, Figure 8

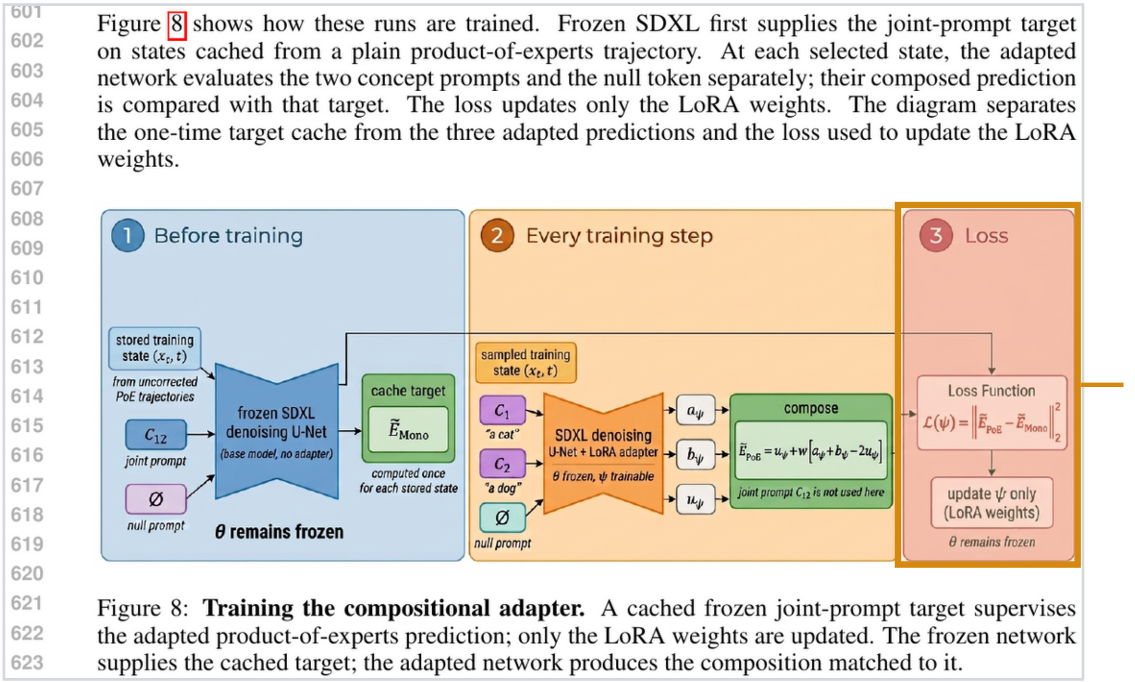


Figure 8 shows how these runs are trained. Frozen SDXL first supplies the joint-prompt target on states cached from a plain product-of-experts trajectory. At each selected state, the adapted network evaluates the two concept prompts and the null token separately; their composed prediction is compared with that target. The loss updates only the LoRA weights. The diagram separates the one-time target cache from the three adapted predictions and the loss used to update the LoRA weights.

1 Before training
stored training state (x_t, t)
from uncorrected PoE trajectories
C_{12}
joint prompt
∅
null prompt
frozen SDXL denoising U-Net
(base model, no adapter)
cache target
$\tilde{E}_{\text{Mono}}$
computed once for each stored state
θ remains frozen
2 Every training step
sampled training state (x_t, t)
C_1
"a cat"
C_2
"a dog"
∅
null prompt
SDXL denoising U-Net + LoRA adapter
θ frozen, ψ trainable
a_ψ
b_ψ
u_ψ
compose
$\tilde{E}_{\text{PoE}} = u_\psi + w[a_\psi + b_\psi - 2u_\psi]$
joint prompt C_{12} is not used here
3 Loss
Loss Function
$\mathcal{L}(\psi) = \|\tilde{E}_{\text{PoE}} - \tilde{E}_{\text{Mono}}\|_2^2$
update ψ only (LoRA weights)
θ remains frozen

Figure 8: **Training the compositional adapter.** A cached frozen joint-prompt target supervises the adapted product-of-experts prediction; only the LoRA weights are updated. The frozen network supplies the cached target; the adapted network produces the composition matched to it.

56173_Restoring_plurality_in_c.pdf (submitted), page 12

the squared error

now reads

replaced by

Equation 12 added

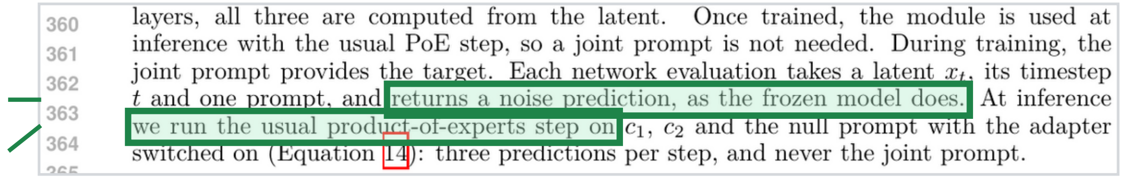
stated in Section 4 instead

THE REVISED ANSWER

Section 4, opening paragraph

returns a noise prediction

inference: PoE, adapter on



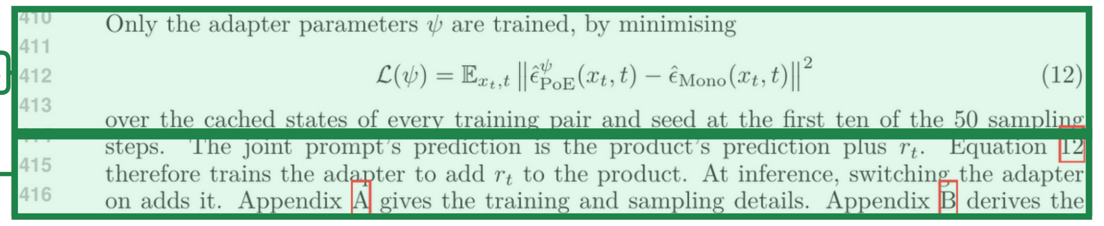
layers, all three are computed from the latent. Once trained, the module is used at inference with the usual PoE step, so a joint prompt is not needed. During training, the joint prompt provides the target. Each network evaluation takes a latent x_t, its timestep t and one prompt, and returns a noise prediction, as the frozen model does. At inference we run the usual product-of-experts step on c_1, c_2 and the null prompt with the adapter switched on (Equation 14): three predictions per step, and never the joint prompt.

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 7

Section 4, Equation 12

Equation 12, the training loss

trains the adapter to add r_t



Only the adapter parameters ψ are trained, by minimising

$$\mathcal{L}(\psi) = \mathbb{E}_{x_t,t}\left\|\hat{\epsilon}^{\psi}_{\text{PoE}}(x_t,t) - \hat{\epsilon}_{\text{Mono}}(x_t,t)\right\|^2 \tag{12}$$

over the cached states of every training pair and seed at the first ten of the 50 sampling steps. The joint prompt's prediction is the product's prediction plus r_t. Equation 12 therefore trains the adapter to add r_t to the product. At inference, switching the adapter on adds it. Appendix A gives the training and sampling details. Appendix B derives the

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 8

The reply we plan to send (replies/k4nd-and-uhz6.md: k4nd W4, its first two parts, and Q4)

answered

W4: method details poorly presented and sometimes inconsistent: The training loss is not stated in Section 4

Section 4 now states the training loss as Equation 12: the squared error between the adapted composition and the joint-prompt prediction, over the cached states of every training pair and seed at the first ten of the 50 sampling steps. The training diagram is removed, since Equation 12 now states the objective in the main text.

W4: method details poorly presented and sometimes inconsistent: Evaluations described as returning the correction

Section 4 now says each adapted evaluation returns a noise prediction, as Appendix B derives. At inference we run the usual product-of-experts step on c_1, c_2 and the null prompt with the adapter switched on: three predictions per step, and never the joint prompt. The joint prompt's prediction is the product's prediction plus the correction r_t, so matching it trains the adapter to add r_t. At inference, switching the adapter on adds it. The reference that printed as "Figure 10" is removed.

Q4: the objective in the main text, and the inconsistencies

The objective is Equation 12 in Section 4. The network outputs and the figure reference are resolved as described under W4. Baseline starting noise is resolved as described under W5.

Changes behind this page: 8ef00a17 Section 4 states the training loss as Equation 12 · 88f02a45 and ead813ab each adapted evaluation returns a noise prediction and inference is the usual PoE step with the adapter on · 2e9af75c the training diagram is hidden.

Question 4 also asks about figure references and starting noise; those are shown on the G9 and G8 pages. The reviewer's quotations are left out of the reply panel because the left column shows them.